Brief History

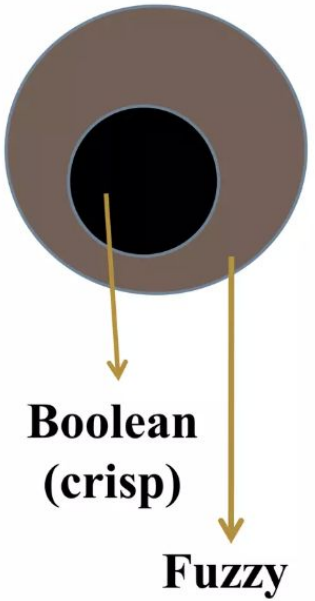
- Classical logic of Aristotle: Law of Bivalence "Every proposition is either True or False(no middle)"

- Jan Lukasiewicz proposed three-valued logic : True, False and Possible

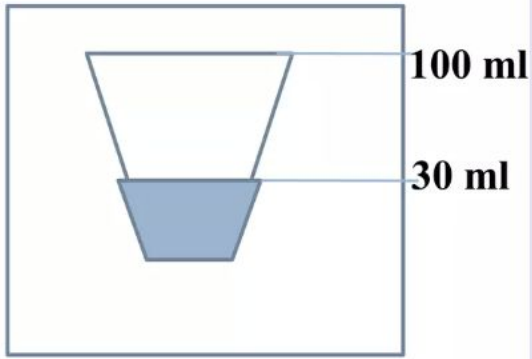
- Finally Lofti Zadeh published his paper on fuzzy logic-a part of set theory that operated over the range [0.0-1.0]

What is Fuzzy Logic?

- Fuzzy logic is a superset of Boolean (conventional) logic that handles the concept of partial truth, which is truth values between "completely true" and "completely false".
- Fuzzy logic is **multivalued.** It deals with **degrees of membership** and **degrees of truth.**
- Fuzzy logic uses the continuum of logical values between 0 (completely false) and 1 (completely true).



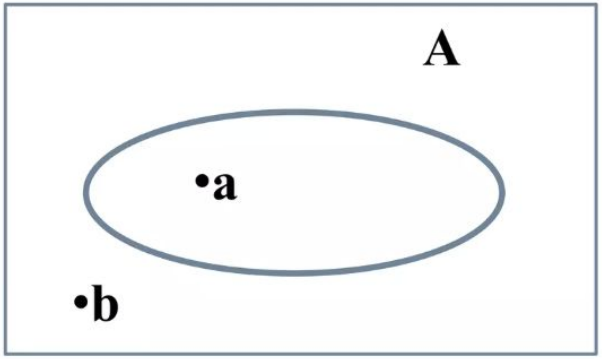
For example, let a 100 ml glass contain 30 ml of water. Then we may consider two concepts: Empty and Full.



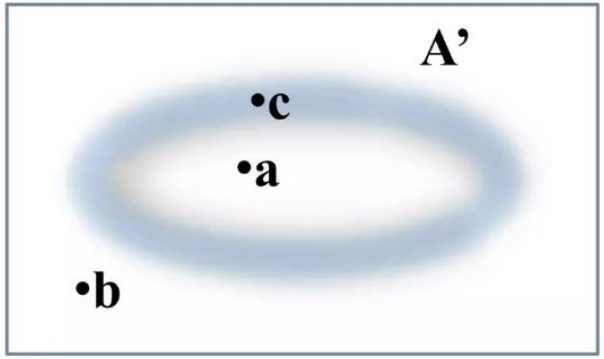
In boolean logic there are two options for answer i.e. either the glass is half full or glass is half empty.

In fuzzy concept one might define the glass as being 0.7 empty and 0.3 full.

Fuzzy Vs. Crisp Set



Crisp set



Fuzzy set

- a: member of crisp set A
- b: not a member of set A

- a: full member of fuzzy set A'
- b: not a member of set A'
- c:partial member of set A'

Fuzzy Vs. Crisp Set

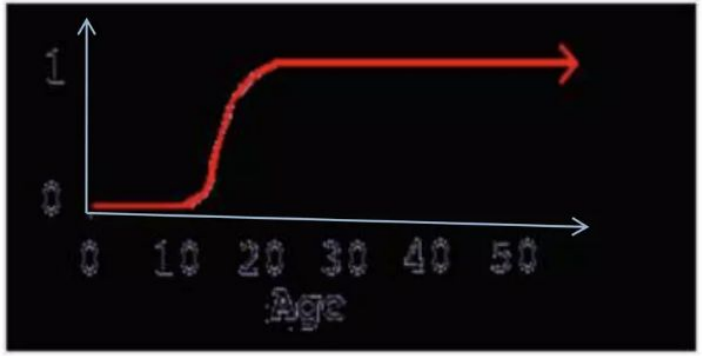
Crisp set

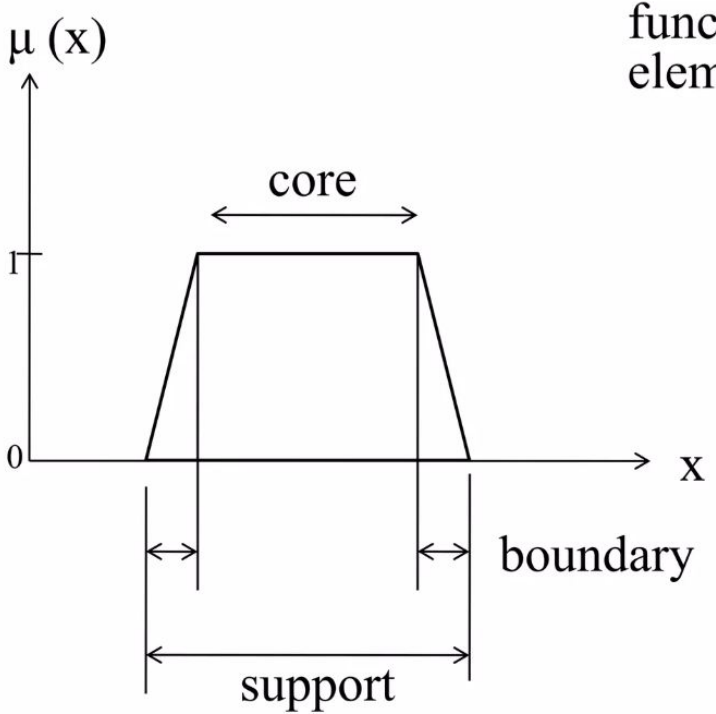
Name	Age	Degree of membership
Sally	5	0
Jenny	18	0
Christen	25	1

Fuzzy set

Name	Age	Degree of membership
Sally	5	0
Jenny	18	0.75
Christen	25	1







A membership function is a mathematical function which defines the degree of an element's membership in a fuzzy set.

- Core: region characterized by full membership in set A' i.e. $\mu(x)=1$.
- Support: region characterized by nonzero membership in set A' i.e. $\mu(x) > 0$.
- Boundary: region characterized by partial membership in set A' i.e. $0 < \mu(x) < 1$

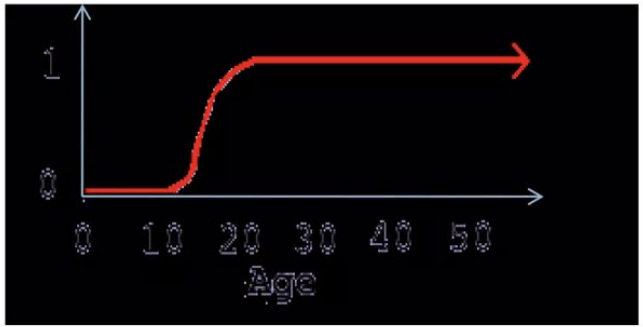
Membership Functions

adult(x)= { 0, if age(x) < 16years

(age(x)-16years)/4, if 16years < = age(x)< = 20years,

1, if age(x) > 20years

}



Fuzzy Logic Vs Probability

- Both operate over the same numeric range and at first glance both have similar values:0.0 representing false(or non-membership) and 1.0 representing true.

- In terms of probability, the natural language statement would be ”there is an 80% chance that Jane is old.”

- While the fuzzy terminology corresponds to “Jane’s degree of membership within the set of old people is 0.80.’

- Fuzzy logic uses truth degrees as a mathematical model of the vagueness phenomenon while probability is a mathematical model of ignorance.

Why use Fuzzy Logic?

- Fuzzy logic is flexible.
- Fuzzy logic is conceptually easy to understand.
- Fuzzy logic is tolerant of imprecise data.
- Fuzzy logic is based on natural language.

2. Crisp and Fuzzy “8”

Crisp 8



Pixel / Region	Membership (μ)
Inside the 8	1
Outside the 8	0

In crisp logic, a pixel either belongs to the set (1) or does not belong to the set (0).

Fuzzy 8



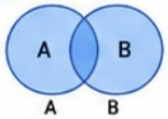
Pixel / Region	Membership (μ)
Center of 8	1.0
Edge of 8	0.6 – 0.8
Near boundary	0.3 – 0.5
Far outside	0.0 – 0.1

In fuzzy logic, a pixel can belong to the set with a degree between 0 and 1.

3. Crisp and Fuzzy Set Operations

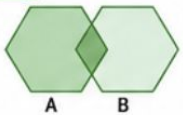
3.1 Disjunction (Union)

Crisp



$A \cup B$ (all elements in A or B)

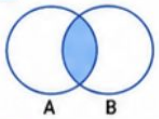
Fuzzy



$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$

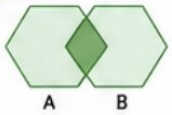
3.2 Conjunction (Intersection)

Crisp



$A \cap B$ (common elements)

Fuzzy



$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$

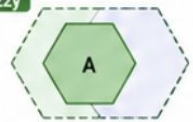
3.3 Complement

Crisp



$\bar{A}$ (not in A)

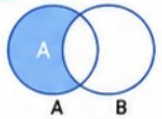
Fuzzy



$\mu_{\bar{A}}(x) = 1 - \mu_A(x)$

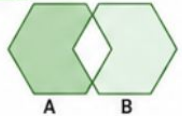
3.4 Difference (A – B)

Crisp



$A - B$ (in A but not in B)

Fuzzy



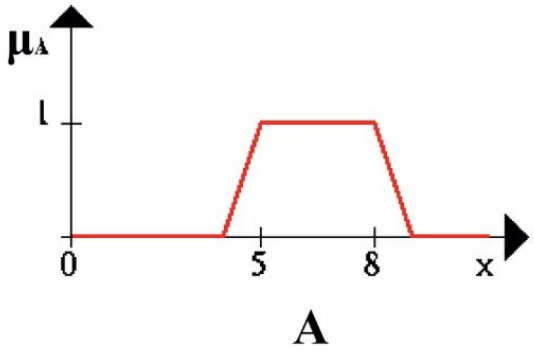
$\mu_{A-B}(x) = \min(\mu_A(x), 1 - \mu_B(x))$

Crisp sets use binary membership (0 or 1) with circles (Venn diagrams), while fuzzy sets use graded membership (0 to 1) with hexagons (for illustration).

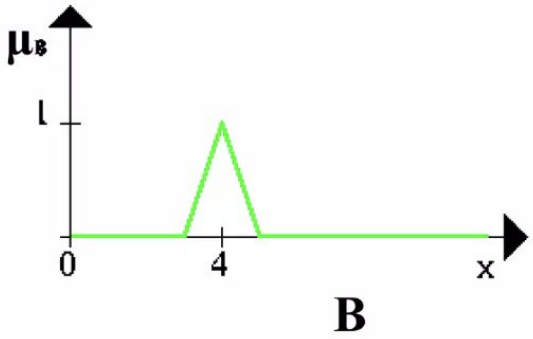
Operations on Fuzzy Set

Consider:

A= {1/2 + .5/3 + .3/4 + .2/5} B= {.5/2 + .7/3 + .2/4 + .4/5}



A



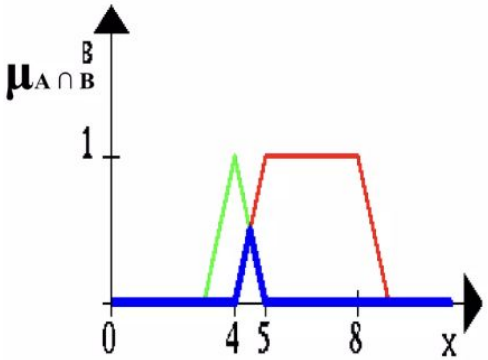
B

>Fuzzy set (A)

>Fuzzy set (B)

>Resulting operation of fuzzy sets

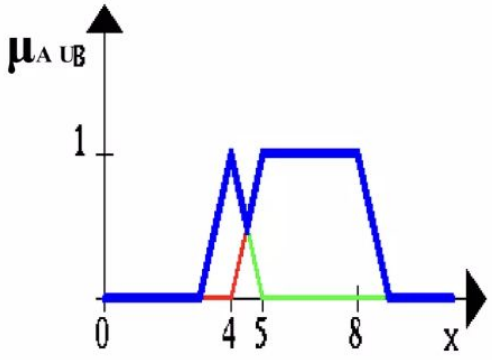
INTERSECTION (A ^ B)



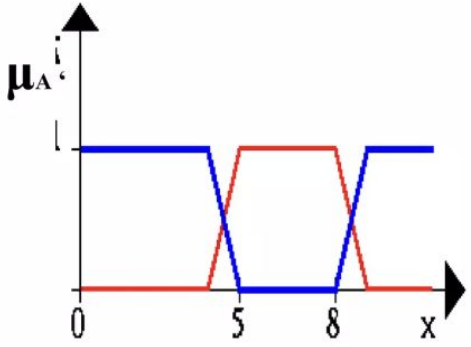
$\mu_{A \cap B} = \min(\mu_A(x), \mu_B(x))$

{.5/2 + .5/3 + .2/4 + .2/5}

UNION (A v B)



COMPLEMENT (¬A)



$\mu_{A'} = 1 - \mu_A(x)$

{1/1 + 0/2 + .5/3 + .7/4 + .8/5}

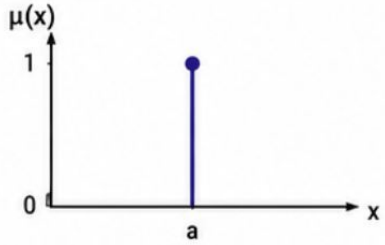
$\mu_{A \cup B} = \max(\mu_A(x), \mu_B(x))$

{1/2 + .7/3 + .3/4 + .4/5}

1. Fuzzification Method

Fuzzification converts a crisp input value into a degree of membership in a fuzzy set using a membership function. Common membership functions are:

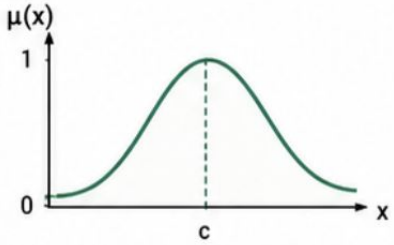
(a) Singleton (Crisp)



$$\mu_A(x) = \begin{cases} 1, & \text{if } x = a \\ 0, & \text{if } x \neq a \end{cases}$$

Example: $a = 30$
(only 30 has membership 1)

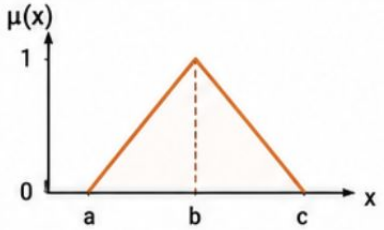
(b) Gaussian



$$\mu_A(x) = \exp\left(-\frac{(x-c)^2}{2\sigma^2}\right)$$

Example: $c = 30$, $\sigma = 5$
(smooth, bell-shaped curve)

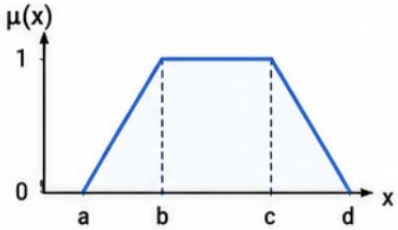
(c) Triangular



$$\mu_A(x) = \begin{cases} \frac{x-a}{b-a}, & a \leq x \leq b \\ \frac{c-x}{c-b}, & b \leq x \leq c \\ 0, & \text{otherwise} \end{cases}$$

Example: $a = 10$, $b = 20$, $c = 30$
(increases then decreases)

(d) Trapezoidal

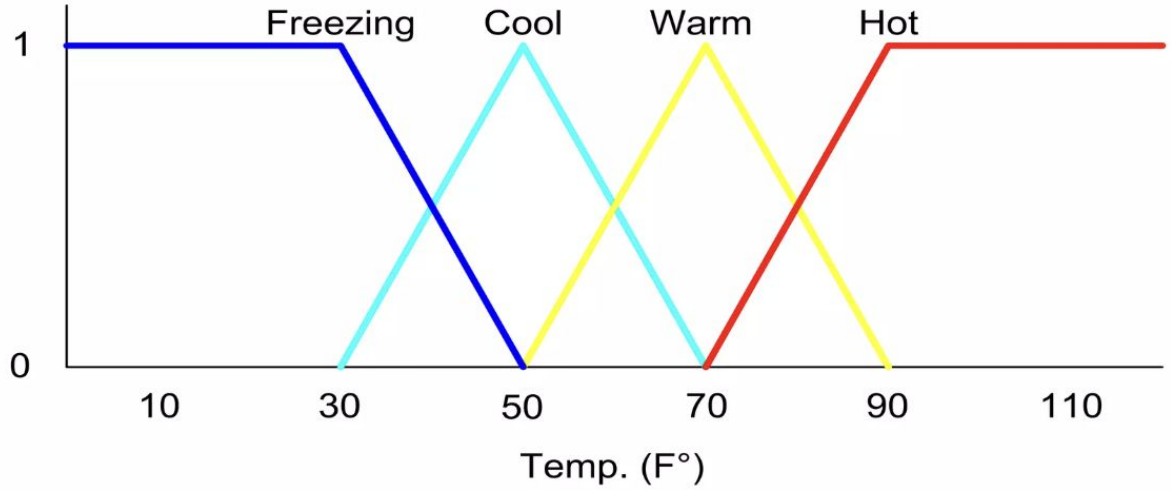


$$\mu_A(x) = \begin{cases} \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & b \leq x \leq c \\ \frac{d-x}{d-c}, & c \leq x \leq d \\ 0, & \text{otherwise} \end{cases}$$

Example: $a = 10$, $b = 20$, $c = 30$, $d = 40$
(flat top region)

Fuzzy Linguistic Variables

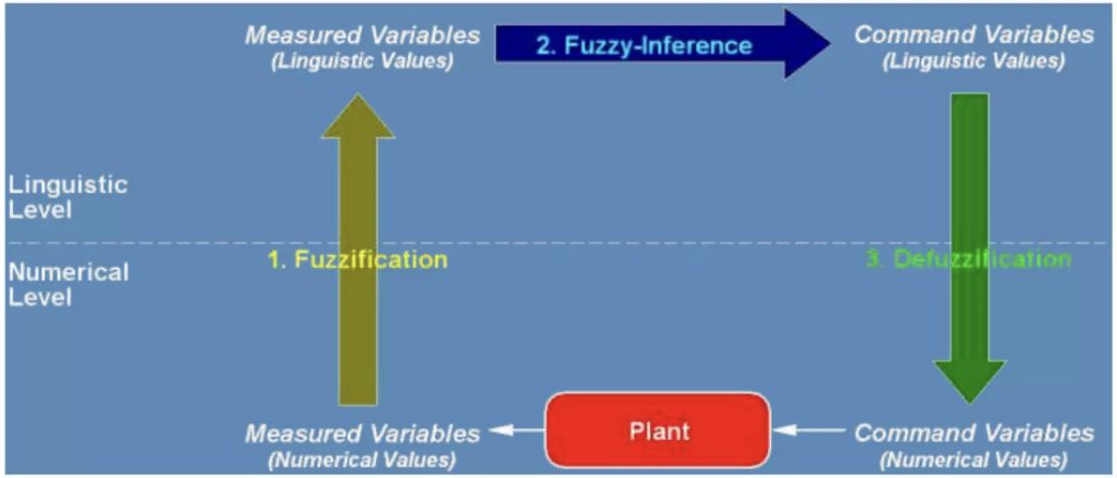
- Fuzzy Linguistic Variables are used to represent qualities spanning a particular spectrum
- Temp: **{Freezing, Cool, Warm, Hot}**



Example Speed Calculation

How fast will I go if it is

- 65 F
- 25 % Cloud Cover ?

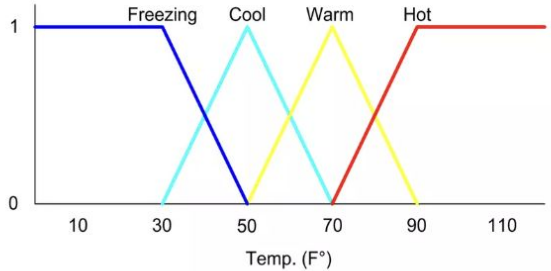


Rules

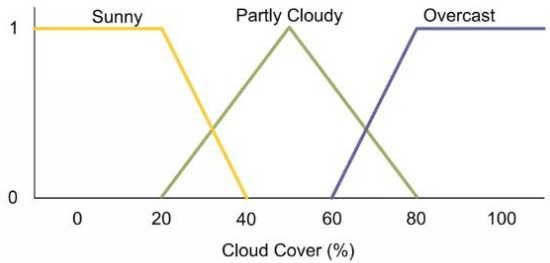
- If it's Sunny and Warm, drive Fast
  $Sunny(Cover) \wedge Warm(Temp) \Rightarrow Fast(Speed)$
- If it's Cloudy and Cool, drive Slow
  $Cloudy(Cover) \wedge Cool(Temp) \Rightarrow Slow(Speed)$
- Driving Speed is the combination of output of these rules...

Input:

Temp: {Freezing, Cool, Warm, Hot}

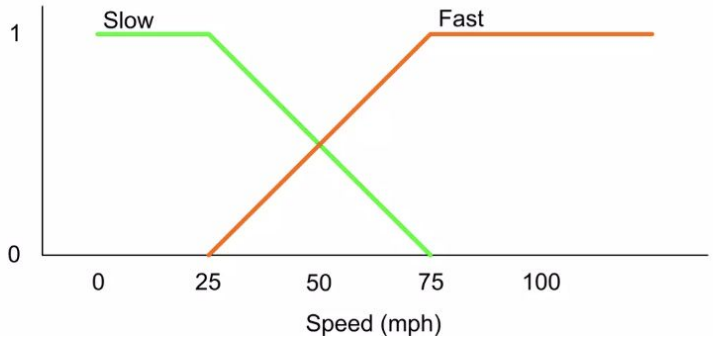


Cover: {Sunny, Partly cloudy, Overcast}



Output:

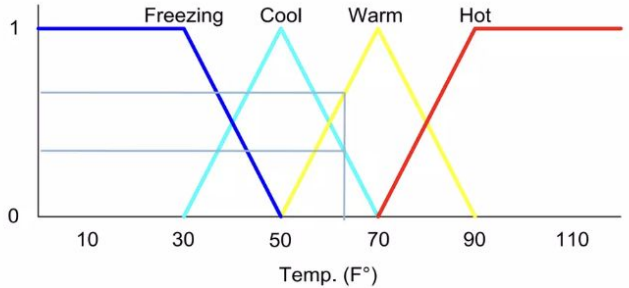
Speed: {Slow, Fast}



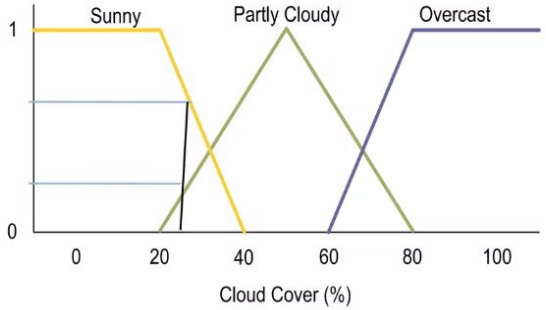
Fuzzification:

Calculate Input Membership Levels

- **65 F ⇒ Cool = 0.4, Warm= 0.7**



- **25% Cover ⇒Sunny = 0.8, Cloudy = 0.2**



Calculating:

- If it's Sunny and Warm, drive Fast

$\text{Sunny(Cover)} \wedge \text{Warm(Temp)} \Rightarrow \text{Fast(Speed)}$

$0.8 \wedge 0.7 = 0.7$

$\Rightarrow$ **Fast = 0.7**

- If it's Cloudy and Cool, drive Slow

$\text{Cloudy(Cover)} \wedge \text{Cool(Temp)} \Rightarrow \text{Slow(Speed)}$

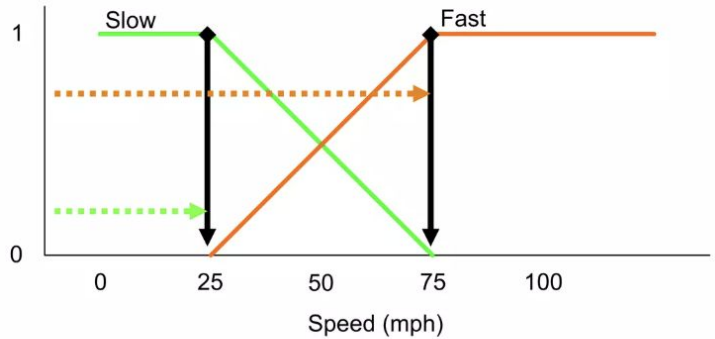
$0.2 \wedge 0.4 = 0.2$

$\Rightarrow$ **Slow = 0.2**

Defuzzification:

Constructing the Output

- Speed is 20% Slow and 70% Fast



- Find centroids: Location where membership is 100%

- Speed = weighted mean
  = (2*25+7*75)/(9)
  = 63.8 mph

Fuzzy Applications

- **Automobile and other vehicle subsystems :** used to control the speed of vehicles, in Anti Braking System.
- **Temperature controllers :** Air conditioners, Refrigerators
- **Cameras :** for auto-focus
- **Home appliances:** Rice cookers , Dishwashers , Washing machines and others

Conclusion

- Fuzzy Logic provides way to calculate with imprecision and vagueness.
- Fuzzy Logic can be used to represent some kinds of human expertise .
- The control stability, reliability, efficiency, and durability of fuzzy logic makes it popular.
- The speed and complexity of application production would not be possible without systems like fuzzy logic.